
VERA System Documentation

Inner Workings, Architecture & Design Decisions

This document explains HOW VERA thinks — the mechanics behind every intelligence feature, data flows between systems, and the reasoning behind key design decisions. For setup and configuration, see README.md.

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1. Design Philosophy

Why Google Apps Script?

Google Apps Script runs in Google's cloud with zero infrastructure cost and no deployment complexity. The alternative — a hosted server (VPS, Lambda, etc.) — would require managing uptime, auth, cron scheduling, and API proxying. Apps Script eliminates all of that:

- **Native Google service access:** `CalendarApp`, `GmailApp`, `DriveApp`, `SpreadsheetApp` are first-class citizens with pre-authorised OAuth. No token management needed.
- **Built-in time-based triggers:** `ScriptApp.newTrigger('nightlyRun').atHour(23).everyDays(1)` replaces an entire cron + server setup.
- **No cold starts for the nightly batch:** The 11pm trigger fires reliably against Google's own infrastructure.
- **Execution logs:** Every `Logger.log()` call is persisted and viewable in the Apps Script execution log — no log aggregation service required.

The trade-offs are real: 6-minute execution time limit, no persistent process, no in-memory caching across executions. These are worked around with Script Properties (persistent key-value store) and per-execution caching via `_configCache_`.

Why Google Sheets as the database?

Google Sheets serves as VERA's operational database. This is a deliberate choice, not a workaround:

- **Visible to the user:** Ahmed can open the sheet, see every flag, task, and metric in a grid, and manually edit any row. There is no hidden state.
- **No schema migrations:** Adding a new column to `TASK_HEADERS` requires updating the constant in `Code.js` and ensuring `ensureSheet()` writes the header on the next run. No SQL ALTER TABLE.
- **Tab-as-namespace:** Each domain (Flags, Tasks, Goals, Itinerary, GymLog, etc.) lives in its own tab. The `TABS` object in `Code.js` defines 70+ tab names — each is effectively a table.
- **Formula-driven dashboards:** The React dashboard reads data via the Web App API, but power users can apply Google Sheets formulas directly to any tab for custom analysis.
- **Audit trail by default:** Sheets preserves edit history natively. Every nightly write is visible in the revision history.

The schema for every tab is defined as a constant array in `Code.js` (e.g. `FLAG_HEADERS`, `TASK_HEADERS`, `GOAL_HEADERS`). The `ensureSheet()` function creates tabs with headers on first run and skips if they already exist — making the system safe to re-run `setupVERA()` without clobbering data.

Why Claude AI for flags (not pure rules)?

Rule-based systems excel at precise conditions: "bill due in 5 days" or "task 14+ days old". But life is ambiguous:

- A meeting labeled "Team Sync" might require Ahmed to prepare a deck — or not. Context determines which.
- A task called "follow up with doctor" means something different if there is an upcoming health appointment on the calendar.
- Spotting that a bucket-list destination is adjacent to an upcoming trip requires geographic reasoning, not a simple date comparison.

Claude reasons over the full assembled context (calendar events, open tasks, summaries, PTO status, interest ledger, bills, home items, goals, trips, financial goals) in a single pass and surfaces what matters. The prompt includes explicit rules for each domain (see Section 2 / `buildPrompt()`) but Claude applies them with judgment, not literal pattern matching.

Rule-based vs AI-assisted split

The split is intentional and deliberate:

Approach	Used for
Rule-based	Escalation aging (3d/7d thresholds), pacing mode miss scoring, pattern recognition triggers, reminder cooldowns, flight status polling, PTO calculation, bill due-date detection
AI-assisted (Claude)	Nightly flag generation, chat responses, packing list generation, weekend planning memos, monthly review synthesis, Explorer daily discovery

Rule-based checks run first, every night, regardless of AI availability. If the Claude API call fails, all the rule-based flags and Slack logs still go out. The Claude call is the only step that can produce zero output on a network error — everything else is local computation.

Why no build pipeline?

The React dashboard (`index.html`) uses Babel Standalone loaded from a CDN. This means:

- No `npm install` , no `webpack` , no build step.
- The file is a single HTML file that can be opened locally, pushed to GitHub Pages, or deployed to Netlify as a static site.
- The entire frontend state is managed with React hooks; no Redux, no router.
- Dark VERA theme is pure CSS variables defined in a `<style>` block.

The trade-off is a slightly longer initial parse time for Babel in the browser. For a personal dashboard used by two people, this is acceptable.

2. The Nightly Intelligence Pipeline

`nightlyRun()` fires at 11pm daily via the Apps Script time-based trigger installed by `setupTriggers()` . It is the orchestration function: it runs every pre-computation step in order, then calls Claude, then writes flags.

All pre-computation steps (Steps -1 through 0q) are wrapped in individual `try/catch` blocks. Failures are pushed into a `stepFailures` array and reported to `#vera-logs` at the end of the run — they do not abort the pipeline. Steps 1–4 (data collection, Claude call, flag write) are inside a single outer `try/catch` ; a failure here sends an error email to `CONFIG.MORNING_NUDGE_EMAIL` and a failure message to `#vera-logs` .

Step -1: `escalateAgedFlags_()`

Scans every unresolved, unacknowledged flag in the `Flags` tab. For each flag:

- If age $\geq$ 3 days and `Escalated` column is empty: bumps urgency up one level (Low→Medium, Medium→High, High stays High) and writes `'3d'` to the `Escalated` column (col K).
- If age $\geq$ 7 days and `Escalated` column is `'3d'`: appends a stale note to the `Reason` field and writes `'7d'` to `Escalated`.

Snoozed flags are skipped while `Snoozed Until` is a future date. Each threshold fires exactly once per flag — the `Escalated` column tracks state.

Step 0: `writeSummarySnapshot()`

Calls `writeMetrics_(ss)` and `writeSummaries_(ss)` in sequence. Both operations clear `[AUTO]`-prefixed rows from their respective tabs and rewrite them fresh. This runs before Claude is called so the AI sees current data.

- `writeMetrics_()` writes task counts, calendar density, and flag counts to the `Metrics` tab.
- `writeSummaries_()` reads `summary_sheet:SourceName` Config rows and pulls live cell values from external Google Sheets.

This step is non-fatal at the outer level but each individual tab write is wrapped internally.

Step 0a: `syncCalendarBirthdaysToImportantDates_()`

Reads upcoming birthday events from the Joint Chaos calendar (events matching a birthday keyword pattern) and writes them to the `Important Dates` tab if not already present. This keeps the People tab's Important Dates populated automatically without manual entry.

Step 0a-ii: `resetChoresByCadence_()`

Scans the `Chores` tab. For each chore, checks whether enough time has elapsed since `Checked At` based on the `Cadence` column (Daily/Weekly/Bi-weekly/Monthly/Quarterly). If the cadence interval has passed, resets `Checked` to false so the chore appears unchecked in the dashboard for the new period.

Step 0b: `writePTOSnapshot_()` → returns `ptoStats`

Reads Ahmed's work calendar (default: `Verizon Calendar`) and classifies all-day events as vacation days, personal time, or company holidays based on title keywords configured in the Config tab. Computes days used, days remaining, burn-rate vs ideal pace, and upcoming travel windows. Writes `[AUTO]` rows to the `Summaries` tab. Returns a `ptoStats` object that is passed downstream to the Monthly Review (Step 0o) and the Claude prompt to avoid re-reading the PTO data twice.

Step 0c: `runExplorer_()`

Calls Claude in "discovery mode" — a separate prompt (built by `buildExplorerPrompt_()`) that asks Claude to surface 2–3 non-obvious, forward-looking observations connecting goals, tasks, calendar, and the Shared Interest Ledger. Uses a date-keyed entry in `Reminders Memory` to fire at most once per day. Output is delivered via `sendNudge_()` — Slack if configured, email otherwise.

Step Od: `getSuppressedKeyPatterns_()`

Reads the `SignalLearning` tab and returns all patterns where `Suppressed = Yes`. The returned array is passed into `generateFlags_()` (Step 2&3) and injected into the Claude prompt as a "SUPPRESSED TOPICS" list — patterns the user has consistently snoozed or ignored will not be re-flagged. See Section 4 for the full scoring logic.

Step Oe: `recordExpiredFlags_()`

Scans the Flags tab for flags that are open (unresolved, unacknowledged, unsnoozed) and either: (a) older than 30 days, or (b) marked `Escalated = '7d'`. For each, calls `recordFlagOutcome_(flagKey, 'expired')` to add a noise-score penalty in Signal Learning. Uses the `Escalated = '7d'` marker as a proxy for "VERA tried, user never responded."

Step Of: `checkPreTripBriefings_()`

Reads the `Itinerary` tab, groups rows by trip key (`YYYY-MM-DD|Trip Label` format), and finds trips whose departure date is within the configured `pretrip_briefing_hours` window (default: 48 hours). For each qualifying trip, assembles a structured briefing flag (via `buildPreTripBriefingFlag_()`) covering weather, packing completion, flight status, itinerary items, and confirmation numbers. Deduplication is handled by `writeFlags_()` — the flag key ensures one briefing per trip, per lifetime.

Step Og: `checkPostTripCapture_()`

Fires `posttrip_capture_delay_days` (default: 1) days after a trip's end date. Writes a flag prompting Ahmed to log trip memories, expenses, and notes. Gated by `posttrip_capture_enabled` Config key.

Step Oh: Morning routine reset

Directly reads the `Morning Routine` tab and sets `Checked = false` and `Checked At = ''` for every item row (columns 5–6). This resets the daily checklist so it starts fresh each morning. Intentionally a direct sheet write, not a function call — it is simple enough to not warrant its own module.

Step Oi: `checkGymSessions_()`

Reads the `Gym Log` tab and the calendar within the configured `gym_tracker_lookback_hours` window (default: 24 hours). Looks for calendar events with titles matching gym/workout patterns that have ended. Writes new rows to `Gym Log` when found. Gated by `gym_tracker_enabled = 'true'` Config key.

Step Oj: `checkFitnessConsistency_()` + `checkFitnessTravelGap_()`

`checkFitnessConsistency()` evaluates gym session count vs the `fitness_weekly_target` Config key. If behind pace by the configured `fitness_low_flag_day` (default Wednesday = day 4), writes a Low-urgency flag. `checkFitnessTravelGap()` checks if a trip is approaching and automatically creates gym session placeholder events in the VERA calendar for the travel period. Both are gated by `fitness_enabled = 'true'`.

Step 0k: `autoRestockItems()` + `generatePantryFlags()`

`autoRestockItems()` uses exponential moving average (EMA, configurable alpha via `pantry_ema_alpha`, default 0.3) to predict when household items will run out based on `Purchase History` tab data, and pre-adds restock items to the Shopping List. `generatePantryFlags()` checks if a trip is approaching and items are predicted to run out during the trip. Both are gated by `pantry_enabled = 'true'`.

Step 0l: `inferCapacityMode()`

Reads tomorrow's calendar and scores load based on meeting count and event density. Writes a capacity mode signal (`busy`, `normal`, or `light`) to Script Properties. The `morningNudge()` function reads this to filter which flags to surface in the morning briefing — `busy` days show only High + time-sensitive flags.

Step 0m: `checkContracts()`

Reads the `Contracts` tab. For each active contract, checks whether the end date is within the notice period (stored in `Notice Period Days` column). Generates Medium flags for contracts approaching their notice window and High flags for expired contracts. Uses source `'General'` and key format `contract_renewal_[name_snake]`.

Step 0n: `checkHealthAppointments()`

Reads Google Calendar for events prefixed with `DR:` (the convention for health appointments). Computes days until the appointment and compares against configured intervals. Generates flags for overdue or upcoming appointments. This replaces the former `HEALTH_APPOINTMENTS` sheet tab — appointment data now lives in the calendar, not a sheet.

Step 0o: `checkMonthlyReview_(ptoStats)`

Fires only on the 1st of each month. Checks whether a review for the prior month already exists in the Flags tab (dedup by key `monthly_review_YYYYMM`). If not, calls `buildMonthlyReview()`, writes a Low-urgency flag, archives to the `Monthly Reviews` tab, and sends a styled HTML email. The `ptoStats` object computed in Step 0b is passed in to avoid re-reading PTO data. See Section 13 for the full assembly.

Step 0p: `resetWeekMealPlan_()` (Saturdays only)

Runs only when `today.getDay() === 6` (Saturday). Archives the current week's meal plan entries and seeds a new week's rows. Gated by `meal_planner_enabled = 'true'`.

Step 0q: `checkCrossPatternFlags_()`

Builds a cross-domain signal snapshot and evaluates 7 compound patterns. See Section 5 for the full pattern definitions and trigger conditions. Writes at most `pattern_max_flags` (default: 2) new flags per run.

Step 1: Data collection

```
events    = getUpcomingEvents()      // Calendar.js - reads all Google Calendars
tasks     = getOpenTasks()           // Tasks.js - reads Tasks tab + Google Tasks
summaries = getSummaries()           // Summaries.js - merges Metrics + Summaries tabs
ledger    = getSharedInterestLedger_() // Interests.js - reads Shared Interests tab
```

`getUpcomingEvents()` reads the number of days configured in `calendar_days_ahead` (default: 7). It reads ALL calendars via `CalendarApp.getAllCalendars()`, applying `skip_calendars` Config rows to exclude noise calendars. Each event gets a `calLabel` from `calendar_label:CalName` Config rows.

Step 1b: `suggestDueDates(tasks)`

For tasks with no due date, Claude suggests a due date based on task content and writes it back to the sheet. This is a separate, non-fatal Claude call made before the main flag generation call.

Step 2: Early exit if no data

If `events.length === 0 && tasks.length === 0 && summaries.length === 0`, the Claude call is skipped entirely. A log entry and a Slack message are sent (`0 flags - no data`). This prevents burning API tokens on empty runs (e.g. quiet travel weekends with no tasks).

Steps 2 & 3: `generateFlags()` — Claude API call

`generateFlags()` in `Claude.js` calls `buildPrompt()` to assemble the full context string, then calls the Anthropic API. The `suppressedPatterns` array from Step 0d is passed in and injected as a "SUPPRESSED TOPICS" block in the prompt. The response is expected to be a raw JSON array — `parseFlags()` strips any markdown fences and JSON-parses the result.

Step 4: `writeFlags(flags)`

Persists Claude's output to the `Flags` tab. See Section 3 for the full dedup and fingerprinting logic.

Step 4b: `recordFlagsGenerated_(flagKeys)`

Extracts the `key` field from each written flag and calls `recordFlagsGenerated_()` in `SignalLearning.js`, incrementing the `Total Seen` counter for each pattern. This feeds the noise-scoring loop described in Section 4.

Slack summary: `sendSlackLog_()`

At the end of `nightlyRun()`, a summary is posted to `#vera-logs`:

✓ Nightly run — 5 flags written (2 High, 2 Med, 1 Low) · 1 step warning in 47s

If `stepFailures` is non-empty, a second message lists each failing step. If the outer try/catch catches a fatal error, it sends a `✗ Nightly run FAILED` message with the error and file/line.

3. Flag System Deep Dive

Flag creation — `writeFlags()`

`writeFlags(flags)` is called from `nightlyRun()`, `checkCrossPatternFlags()`, `checkPreTripBriefings()`, `checkMonthlyReview()`, `checkMissRate()`, `EmailParser.js`, and other rule-based checkers. It is the single write path for all flags.

Each flag object must contain: `source`, `flag`, `reason`, `urgency`, and optionally `key`.

Fingerprinting and dedup: Before writing, `writeFlags()` builds a `Set` of fingerprints from ALL existing flags (regardless of state — open, acknowledged, snoozed, or resolved). Once a key has been written in any state, it is permanently blocked from re-appearing. If the issue genuinely recurs, Claude is expected to generate a new key (e.g. adding a `_q2` suffix).

The fingerprint is computed by `makeFlagFingerprint(source, flagText, key)`:

- If a stable `key` is present: fingerprint = `'key:' + safeKey` (key alone, no source prefix)
- If no key (legacy rows): fingerprint = `source + '|' + first 8 words of flag text`

Fuzzy dedup via `keysAreSimilar()`: For key-bearing flags, checks whether the new key shares $\geq 60\%$ of its meaningful tokens with any existing key fingerprint in the set. Month names and standalone numbers are stripped before comparison, so `verizon_bill_march_13` and `verizon_bill_march_14` are treated as the same issue. This prevents date-drifted duplicates when a recurring bill flag shifts by one day.

Flag ID format: `FLAG-YYYYMMDD-NN` where `NN` is a random two-digit number (10–99) chosen at write time. The randomness prevents collision when multiple flags are written in the same nightly run.

Color coding: After writing, `colorCodeFlags()` applies background colors to all rows in the Flags tab:

- High = `#fce8e6` (soft red)
- Medium = `#fef9e7` (soft yellow)
- Low = `#e6f4ea` (soft green)

Flag key convention

Keys are stable, snake_case identifiers designed by Claude. The convention is `domain_description_context`, for example:

- `health_appt_ahmed_annual_physical`

- `cross_high_stress_compound`
- `verizon_bill_march_13`
- `task_neglect_TASK_0012`
- `fgoal_emergency_fund_at_risk`
- `packing_not_started_austin_trip`

Keys serve as the cross-run dedup anchor. When Claude generates the same conceptual flag on consecutive nights (e.g. a bill that is still unpaid), the key should match the existing flag in the sheet, causing `writeFlags()` to skip it as a duplicate.

Escalation — `escalateAgedFlags_()`

Escalation runs at the very start of `nightlyRun()` (Step -1) so the urgency of aged flags is correct before the morning nudge reads them. The `Escalated` column (col K, index 10) stores `'3d'` and `'7d'` as watermarks so each threshold fires exactly once per flag lifetime. Snoozed flags are excluded from escalation while the snooze date is in the future.

Flag lifecycle

```

created (Open)
  ↓
acknowledged (Acknowledged = 'Yes') OR
snoozed (Snoozed Until = future date) OR
resolved (Resolved = 'Yes')

```

Snoozed flags re-surface automatically: `writeFlags()` does not block them, and `escalateAgedFlags_()` skips them while snoozed — but the flag is still visible in the sheet and the dashboard. When the snooze date passes, the flag becomes active again.

When a user resolves a flag (via the dashboard or Slack button), `recordFlagOutcome_(flagKey, 'resolved')` is called, adding a positive signal score to the pattern in Signal Learning.

4. Signal Learning & Noise Filtering

Signal Learning tracks how Ahmed engages with each flag pattern over time and uses that history to suppress recurring noise.

`SignalLearning` tab structure

Headers: `Key Pattern | Total Seen | Acknowledged | Snoozed | Resolved | Expired/Ignored | Last Seen | Score | Suppressed`

The `Key Pattern` column stores the output of `extractKeyPattern()`, which strips date and sequence suffixes from full flag keys:

- `verizon_bill_20260315` → `verizon_bill`
- `task_neglect_TASK_0012` → `task_neglect`
- `ergonomic_break_1234` → `ergonomic_break`

This normalization is critical: it means all variants of a recurring pattern accumulate into one row rather than spreading across dozens.

Score formula

```
score = 100 - (snoozed × 20) - (expired × 15) + (ack × 10) + (resolved × 25)
score = clamp(0, 100)
```

- **Snoozed (-20):** User acknowledged the flag was surfaced but chose to defer — mild negative signal.
- **Expired/Ignored (-15):** Flag sat open > 30 days with no action — silent dismissal.
- **Acknowledged (+10):** User read it and marked it done — mild positive signal.
- **Resolved (+25):** User took action and resolved the underlying issue — strong positive signal.

Suppression threshold

A pattern is marked `Suppressed = Yes` when: `score < 25` AND `total_seen ≥ 5`.

The minimum sightings gate (5) prevents suppression from triggering on a pattern that was snoozed once or twice early on. The score threshold (25) means a pattern needs to be consistently ignored or snoozed multiple times before VERA stops surfacing it.

How suppression flows into the pipeline

1. **Step 0d** (`getSuppressedKeyPatterns()`) reads the `SignalLearning` tab at the start of `nightlyRun()` and returns all suppressed patterns.
2. The array is passed into `generateFlags()` → `buildPrompt()` → injected as a "SUPPRESSED TOPICS" block: *"user has consistently indicated these are not useful — do NOT generate flags for them."*
3. **Anticipator rules** (Reminders.js) check `getSuppressedKeyPatterns()` before sending nudges for bills due, packing reminders, goal check-ins, and home service items.

Feedback loop

- `recordFlagsGenerated_(flagKeys)` is called after each `writeFlags()` call — increments `Total Seen` for each pattern key.
- `recordExpiredFlags_()` is called each night (Step 0e) — increments `Expired/Ignored` for flags open > 30 days.
- `recordFlagOutcome_(flagKey, outcome)` is called when a user acknowledges, snoozes, or resolves a flag via the dashboard or Slack.

`getTopSignalPatterns_(n)` returns the top N patterns by score (highest first). These are injected into the Claude prompt as "SIGNAL LEARNING — what Ahmed engages with most" to bias flag priority toward topics he cares about.

5. Cross-Domain Pattern Recognition

`PatternRecognition.js` implements meta-intelligence: compound signals that span multiple VERA data domains. Single-domain flags (one overdue task, one missed gym session) are left to their individual checkers. The Pattern Recognition engine only fires when two or more domains signal simultaneously.

`buildCrossDomainSnapshot_()`

Assembles a lightweight signal object from existing VERA data functions. Every data read is wrapped in its own `try/catch`: if a domain is unavailable (e.g. Gym Log tab empty), that field is set to `null` and any pattern requiring it is skipped gracefully. The snapshot includes:

Field	Source
<code>overdueTaskCount</code>	<code>getOpenTasks()</code> — tasks where <code>isOverdue = true</code>
<code>neglectedTaskCount</code>	<code>getOpenTasks()</code> — tasks where <code>ageInDays ≥ 14</code>
<code>calendarEventsThisWeek</code>	<code>getUpcomingEvents()</code> — events in next 7 days
<code>activeGoals</code>	<code>getGoals_()</code> — goals with status <code>'Doing'</code>
<code>gymSessionsThisWeek</code>	<code>getGymLog_()</code> — sessions attended in last 7 days
<code>gymTargetPerWeek</code>	Config key <code>gym_sessions_per_week</code> (default 3)
<code>overdueHealthAppts</code>	<code>getHealthAppointments_()</code> — <code>daysUntil < 0</code> and <code>intervalMonths ≠ 0</code>
<code>activeHighFlags</code>	Direct Flags tab read — unacknowledged, unresolved, urgency = <code>'High'</code>
<code>pacingMode</code>	<code>isInPacingMode_()</code>
<code>vacationMode</code>	<code>isInVacationMode_()</code>
<code>intensityLevel</code>	<code>computeIntensitySignal_()</code> — see Section 6
<code>takeoutRatio</code>	<code>getMealPlanHistory_(2)</code> — takeout/eating-out fraction of meals in past 7 days

`evaluateCrossPatterns_(snap)` — the 7 patterns

All patterns null-check their required fields before evaluating. A pattern whose required fields are null is silently skipped.

Pattern 1: High-Stress Compound (`cross_high_stress_compound` , urgency: High)

- Trigger: `overdueTaskCount ≥ 3` AND `activeHighFlags ≥ 2` AND `gymSessionsThisWeek == 0`
- Rationale: tasks are piling up, high-urgency flags are unresolved, and the user isn't getting physical release — a compounding stress signal.

Pattern 2: Goal–Behaviour Drift (`cross_goal_behaviour_drift` , urgency: Medium)

- Trigger: `activeGoals ≥ 2` AND `gymSessionsThisWeek < gymTargetPerWeek * 0.5` AND `overdueTaskCount ≥ 2`
- Rationale: stated goals are active but execution is stalling on both fitness and tasks — declared intent is diverging from actual behaviour.

Pattern 3: Social/Calendar Gap (`cross_social_calendar_gap` , urgency: Low)

- Trigger: `calendarEventsThisWeek ≤ 1` AND `pacingMode == false` AND `vacationMode == false`
- Rationale: an unusually empty week that is not intentional (not pacing mode, not vacation) — may indicate social drift or missed opportunity for connection.

Pattern 4: Overload + Pacing Mismatch (`cross_overload_no_pacing` , urgency: Medium)

- Trigger: `intensityLevel == 'high'` AND `pacingMode == false` AND `activeHighFlags ≥ 2` AND `overdueTaskCount ≥ 3`
- Rationale: the system signals high load but the user hasn't activated pacing mode — they are running hard without a buffer.

Pattern 5: Meal Chaos (`cross_meal_chaos` , urgency: Low)

- Trigger: `takeoutRatio ≥ 0.7` AND `overdueTaskCount ≥ 2`
- Rationale: mostly takeout combined with task pile-up is a compound signal of a chaotic life period, not just a food preference.

Pattern 6: Backlog Accumulation (`cross_backlog_accumulation` , urgency: Medium)

- Trigger: `neglectedTaskCount ≥ 5` AND `calendarEventsThisWeek ≥ 3`
- Rationale: the calendar is full but tasks aged 14+ days are building up — the user is attending meetings but not processing their backlog.

Pattern 7: Health Neglect Compound (`cross_health_neglect` , urgency: Medium)

- Trigger: `overdueHealthApts ≥ 2` AND `gymSessionsThisWeek == 0`
- Rationale: multiple overdue appointments combined with no gym sessions this week is a compound health-neglect signal.

Deduplication and rate limiting

Candidates are sorted High → Medium → Low. Before writing, the function reads all existing `Pattern Recognition` source flags and builds a `suppressedKeys` map: a pattern key is suppressed if it has an open (unresolved) flag, or if it was resolved within the last `pattern_dedup_days` (default: 7) days. A maximum of `pattern_max_flags` (default: 2) new flags are written per run. Both values are overridable via Config tab keys `pattern_max_flags` and `pattern_dedup_days` .

6. Pacing & Capacity Mode

`computeIntensitySignal_(activeFlags, tasks, events)`

Defined in `WeekendPlanner.js` but used by both the Weekend Planner and `PatternRecognition.js`.

Evaluates three thresholds:

- Active unresolved flags $\geq 4 \rightarrow$ threshold breach
- Overdue tasks $\geq 2 \rightarrow$ threshold breach
- Meetings in the next 5 days (non-all-day events) $\geq 8 \rightarrow$ threshold breach

`level = 'high'` if 2+ thresholds breached, `'medium'` if 1 threshold breached, `'low'` otherwise.

Returns `{ level, activeFlagCount, overdueTaskCount, meetingCount }`.

`isInPacingMode_()`

Fast read from Script Properties. Reads `PACING_MODE_ACTIVE`. Auto-expires: if `today > PACING_MODE_ENDS` date, sets `PACING_MODE_ACTIVE = 'false'` and returns false. Called from `PatternRecognition.js`, `Fitness.js`, and `Reminders.js`.

`isInVacationMode_()`

Fast read: returns `true` if `VACATION_MODE_ACTIVE === 'true'` in Script Properties. Set by `checkVacationMode_()` which runs inside `checkPacing_()` at the start of each nightly run.

`checkVacationMode_()` — vacation detection logic

Reads the `Itinerary` tab to build a date-range map (`tripKey  $\rightarrow$  { min, max }`). Reads `TripMeta` to build a traveler map (`tripKey  $\rightarrow$  traveler string`). Finds the first active trip where today falls within the date span and the trip is not Victoria-only (a trip is Victoria-only if the traveler string contains `'victoria'` and does not contain `'ahmed'`). Writes `VACATION_MODE_ACTIVE`, `VACATION_MODE_ENDS`, and `VACATION_TRIP_NAME` to Script Properties. Fires a Memory Log event on first detection and on transition back to inactive.

`checkMissRate_()` — the pacing offer loop

Scores 4 domains for misses in the last 48 hours:

1. **Gym:** any scheduled session today or yesterday with `attended  $\neq$  'Yes'`
2. **Tasks:** any Open task with a due date of today or yesterday
3. **Flags:** $\geq$ `pacing_flag_threshold` (default 3) unacknowledged Medium/High flags open
4. **Chores:** any chore whose last `Checked At` + cadence interval has elapsed

If 2+ domains hit:

- **First detection:** Writes a Medium flag offering to activate pacing mode. Stores `PACING_FIRST_DETECTED` (timestamp) and `PACING_OFFER_FLAG_KEY` in Script Properties.
- **48 hours later (still 2+ domains hit, offer flag unacknowledged):** Auto-activates pacing mode. Defers all upcoming non-recurring open tasks with a future due date to next Monday. Sets `PACING_MODE_ACTIVE = 'true'` and `PACING_MODE_ENDS = today + pacing_mode_days` (default 7 days). Writes a Low flag confirming activation.

Pacing mode effects: Pattern 4 (Overload+Pacing Mismatch) and Pattern 1 (High-Stress Compound) suppress their flags while pacing mode is active. The Anticipator's hydration, ergonomic, and mobility rules still fire — only contextually inappropriate nudges are suppressed.

7. Anticipator — Reminder Engine

`hourlyCheck()` fires every hour via the Apps Script `everyHours(1)` trigger installed by `setupTriggers()`. Apps Script does not support conditional hour windows in trigger configuration — the function itself contains all guard logic.

`runAnticipatorRules(now, hour, isWeekday, cfg)`

Runs 8 rules in sequence. Each rule is wrapped in its own `try/catch` — a failure in one rule does not block the others.

Rule	When it fires	Cooldown
<code>checkErgonomicBreak_</code>	Weekdays 9am–6pm	55 min (Slack) / 180 min (email)
<code>checkHydration_</code>	Weekdays 8am–6pm	110 min (Slack) / 180 min (email)
<code>checkCalendarOpportunity_</code>	Weekdays 9am–5pm	90 min (Slack) / 180 min (email), once per date+hour key
<code>checkEveningMobility_</code>	Configured hour (default 8pm)	1440 min (once per day)
<code>checkBillsDue_</code>	Daily 8–9am	1440 min per bill per day
<code>checkTripPackingReminder_</code>	Daily 8–9am	1440 min per trip per day
<code>checkGoalCheckin_</code>	Mondays at <code>weekend_planner_hour</code> (default 8am)	10080 min (once per week)
<code>checkHomeServiceDue_</code>	Daily 8–9am	1440 min per item per day

Reminders Memory `tab ( TABS.REMINDERS_MEMORY )`

Every sent nudge is logged to this tab with columns: `Rule Key` | `Sent At` | `Message` (first 100 chars).

`wasRecentlySent_(ruleKey, cooldownMinutes)` scans from the bottom (most recent first) for the most recent

entry matching the rule key. If the elapsed time since that entry is less than `cooldownMinutes`, the rule is blocked.

Suppression integration

`checkBillsDue()`, `checkTripPackingReminder()`, `checkGoalCheckin()`, and `checkHomeServiceDue()` each call `getSuppressedKeyPatterns()` before firing. If the pattern key for that bill, trip, or home item is suppressed, the nudge is skipped entirely.

`sendNudge(ruleKey, subject, message)`

Delivery logic:

1. If `isSlackConfigured()` is true (i.e. `SLACK_BOT_TOKEN` is set in Script Properties): sends to `#vera-notifications`.
2. Else: sends a plain-text email via `MailApp.sendEmail()` to `CONFIG.MORNING_NUDGE_EMAIL`.

Always calls `markSent(ruleKey, message)` after delivery to record the send in `Reminders Memory`. Also calls `sendSlackLog()` to post a delivery confirmation to `#vera-logs`.

Weekend Planner integration

`hourlyCheck()` contains an additional gate: `if (day === 1 && hour === cfg['weekend_planner_hour'])`, which calls `runWeekendPlanner()`. This fires every Monday at the configured hour (default 8am). The planner uses a 9000-minute cooldown in `Reminders Memory` (approximately 6.25 days) to prevent double-sends.

`runWeekendPlanner()` assembles clear weekend windows (via `findClearWindows()` filtered to windows that touch a Friday or Monday), computes the intensity signal, reads goals, the Shared Interest Ledger, and PTO stats, then calls Claude to generate a "Weekend Decision Memo" with three archetypes:

- **THE EXTENSION:** goal-anchored activity
- **THE CONTRAST:** rest/recharge (weighted heavier if intensity is high)
- **THE PROTOTYPE:** new experience not in the Interest Ledger

The memo is delivered via `sendNudge()` and also written as an all-day Google Calendar event on the upcoming Saturday in the `pto_vera_calendar` (default: `'Vera'` calendar).

8. Chat System Architecture

Context assembly

Every chat turn in `Chat.js` assembles a fresh context object via `buildChatContext()`. The context includes:

- Active flags (unresolved, unacknowledged) from the Flags tab
- Open tasks from `getOpenTasks()`
- Upcoming events from `getUpcomingEvents()`
- Life summaries from `getSummaries()`
- Bills from the Bills tab
- Home items from the Home Items tab
- Goals from `getGoals_()`
- Financial goals
- Projects
- Travel context (upcoming trips, packing items by trip key, itinerary items by trip key)
- Career position, goals, progression, development, wins, network
- And more domain-specific data depending on the chat turn context

This context is assembled fresh on every turn — there is no persistent in-memory state between chat calls (Apps Script does not support persistent processes).

`buildChatSystemPrompt_(context)`

The system prompt injects the assembled context as structured sections for Claude to reason over. It also injects:

- Proactive insights from `computeProactiveInsights_(context)` — surfaced as "VERA NOTICES" in the prompt
- Source routing identity (Ahmed vs Victoria vs Slack)
- Persona definition and action capabilities

Proactive insights — `computeProactiveInsights_(context)`

Pre-computes from context data (no additional API calls) and returns a formatted `[HIGH/MEDIUM/LOW]` notice string. Sources checked:

1. Tasks overdue > 3 days (by name, sorted by staleness, top 3)
2. Bills due within 7 days and unpaid (urgency: `≤ 2 days = HIGH` , `≤ 5 = MEDIUM` , `≤ 7 = LOW`)
3. Upcoming trips within 14 days: empty packing list (`HIGH` if within 7 days), unpacked items (`MEDIUM`), empty itinerary (`MEDIUM`)
4. Home items with service overdue (`HIGH`) or due within 7 days (`LOW`)
5. Projects with overdue tasks (`MEDIUM`)

These are injected into the system prompt so Claude can surface them proactively at the start of a conversation or when contextually relevant.

Source routing

The `body.source` field determines how Claude identifies the conversation participant:

- `'dashboard'` : Ahmed (primary user)
- `'dashboard-lite'` : Victoria (simplified dashboard)
- `'vera-chat'` : Slack `#vera-chat` channel — identity derived from Slack user ID via `getSlackUserName_()`

Action dispatch

Chat.js contains a large `dispatchAction_()` function that handles Claude's structured action responses. Claude returns a JSON action object (`{ action, params }`) alongside its conversational reply. Supported actions include (but are not limited to): `add_task` , `complete_task` , `resolve_flag` , `snooze_flag` , `acknowledge_flag` , `update_goal` , `add_trip_item` , `add_packing_item` , `log_gym_session` , `add_bill` , `add_home_item` , `update_project_task` , `record_purchase` , and many more. Each action modifies the appropriate sheet tab directly.

Multi-turn context

Chat history is stored per session in Script Properties under the key `CHAT_HISTORY_{sessionId}` as a JSON-serialised array of `{ role, content }` messages. The `CHAT_MAX_EXCHANGES` constant (default: 10) limits the stored back-and-forth to 20 messages (10 pairs). Older messages are trimmed from the front of the array.

Web search

When `VERA_SEARCH_API_KEY` is configured in Script Properties, the `WEB_SEARCH_TOOL_` is appended to the Claude API call's `tools` array. Claude can invoke it for real-time or location-specific queries. A PII scrub (email, phone, SSN regex) runs on every query before it leaves Apps Script. The engine is configurable via `VERA_SEARCH_ENGINE` Script Property (`'serper'` or `'tavily'` , default `'serper'`).

9. Slack Integration Architecture

Inbound (user → VERA)

Slack sends `POST` requests to the Apps Script web app `doPost()` endpoint via the Events API. Two payload shapes arrive:

1. **JSON body** (`body.type == 'event_callback'`): routed to `handleSlackEvent_()` in `Slack.js`
2. **Form-encoded body** (button interactions, slash commands): routed to `handleSlackFormPost_()` in `Slack.js`

Apps Script does not reliably decode form-encoded POST bodies into `e.parameter` . `handleSlackFormPost_()` falls back to manually URL-decoding `e.postData.contents` when `e.parameter` is empty.

Bot message filtering: `handleSlackEvent()` checks `event.bot_id` and `event.subtype === 'bot_message'` before processing. VERA's own outbound messages arrive back via the Events API and must be ignored to prevent echo loops.

Slack user allowlist: `getSlackAllowedUserIds()` reads `SLACK_ALLOWED_USER_IDS` from Script Properties (comma-separated Slack user IDs). Only allowlisted users can trigger VERA via Slack. Unknown users receive a silent rejection.

Async queuing: Chat messages from `#vera-chat` are queued in `CacheService` and processed asynchronously. This is required to respond within Slack's 3-second acknowledgement deadline — the Apps Script execution (which involves calling Claude) can take 10–30 seconds.

Outbound (VERA → user)

Three send functions handle outbound messages:

- `sendSlackMessage(channelId, text, blocks, threadTs)` : Generic send to any channel. Accepts optional Block Kit `blocks` array and optional `threadTs` for threading.
- `sendSlackNotification(text, blocks, urgency)` : Routes to `#vera-notifications` (`SLACK_NOTIFICATIONS_CHANNEL_ID`). For `urgency === 'High'`, prepends `<@AHMED_USER_ID>` to trigger a Slack mention.
- `sendSlackLog(text)` : Plain text to `#vera-logs` (`SLACK_LOGS_CHANNEL_ID`). Called by `nightlyRun()`, reminder delivery, pattern recognition, travel pipeline, and monthly review.

Block Kit

`buildFlagBlocks(flag)` assembles a three-block structure for each flag:

1. `section` block with urgency emoji + bold flag title + italic reason (truncated to 200 chars)
2. `actions` block with two buttons: `acknowledge_flag` (`action_id`) and `snooze_flag` (`action_id`), each carrying the flag ID as the `value`
3. `divider` block

When a user clicks Acknowledge or Snooze, Slack sends an interaction payload (form-encoded JSON) to `doPost()`. `handleSlackInteraction()` parses the action ID and value, updates the flag in the sheet, and calls `sendSlackResponse(responseUrl, confirmText, null, true)` with `replace_original: true` to replace the button message with a confirmation.

The morning nudge uses `buildMorningNudgeBlocks()` which groups flags by urgency (High → Medium → Low) and includes a capacity mode ticker and a "held from yesterday" count.

Channel routing

Channel	Script Property key	Purpose
<code>#vera-chat</code>	<code>SLACK_CHAT_CHANNEL_ID</code>	Bidirectional conversation (Ahmed + Victoria)

#vera-notifications	SLACK_NOTIFICATIONS_CHANNEL_ID	Flag alerts with Acknowledge/Snooze buttons
#vera-logs	SLACK_LOGS_CHANNEL_ID	System event log — nightly run, feature fires, errors

10. Travel Pipeline

Trip detection

Trips are identified by the `Itinerary` tab. Each row has a `Trip Key` column with the format `YYYY-MM-DD|Trip Label` (e.g. `2026-06-15|Nashville Weekend`). The departure date is the `YYYY-MM-DD` prefix. The `TripMeta` tab stores per-trip context: a `traveler` field identifies which trips are Ahmed-only, Victoria-only, or joint. Victoria-only trips are excluded from `checkVacationMode()` and the pacing system.

Email parser — `runEmailScan()`

Runs every 30 minutes via the `runEmailScan` trigger (only when `email_parser_enabled = 'true'` in Config). The process:

1. Runs a broad Gmail search via `buildTravelSearchQuery()` — returns up to 50 threads
2. Deduplicates against the `Processed Emails` tab (`isAlreadyProcessed(messageId)`)
3. Batches new candidates (up to `BATCH_SIZE = 20`) to Claude with a confidence scoring prompt
4. Per-email response: `confidence ≥ HIGH_CONF (0.85)` → auto-process (write/enrich itinerary row);
`LOW_CONF (0.60) ≤ confidence < HIGH_CONF` → hold and write a flag for manual confirmation;
`confidence < LOW_CONF` → discard silently
5. Marks every processed email in `Processed Emails` tab with outcome and mode

The email body is truncated to `MAX_SIGNAL_CHARS = 2500` characters before sending to Claude. The warning in `setupTriggers()` notes that with emails per day up to the 50-thread search limit and 30-minute intervals, this could generate up to 144 Claude API calls per day — the config key `email_parser_enabled` defaults to `'false'`.

Pre-trip briefing — `checkPreTripBriefings()`

Reads the `Itinerary` tab, finds trips departing within `pretrip_briefing_hours` (default: 48 hours), and assembles a `buildPreTripBriefingFlag()` for each. The briefing flag is written as `urgency: 'High'` with source `'Trip Briefing'`. Deduplication via `writeFlags()` key fingerprinting ensures one briefing per trip, ever — the key includes the trip key, so it fires once and never again even if the nightly run repeats. Logged to `#vera-logs` : `:airplane: Pre-trip briefing written - Nashville Weekend`.

Flight status monitor — `checkFlightStatuses()`

Polls the AviationStack API every 15 minutes via the `checkFlightStatuses_` trigger. Active only for flights within 24 hours of departure. Polling intervals are rate-limited:

- Departure > 6h away (in the 6–24h window): every 3 hours
- Departure 1–6h away: every 60 minutes
- Departure < 1h away: every 15 minutes

Status is stored in: (a) the Itinerary tab row's `Metadata` column (JSON with `flight_status` key) for sheet-sourced flights; (b) Script Properties under `FLIGHT_STATUS_CACHE` (keyed by `CAL-xxx` IDs) for calendar-sourced flights.

When a 429 response is received from AviationStack: a monthly quota error backs off for 30 days; a rate-limit error backs off for 2 hours. The backoff window is stored in `AVIATIONSTACK_BACKOFF_UNTIL` Script Property and checked at the top of every poll.

Post-trip capture — `checkPostTripCapture_()`

Fires `posttrip_capture_delay_days` (default: 1) after a trip's end date. Writes a flag (source: `'Trip'`) prompting Ahmed to log memories, expenses, and notes. Gated by `posttrip_capture_enabled = 'true'`.

11. PTO & Clear Windows

PTO snapshot — `writePTOSnapshot_()`

Reads Ahmed's work calendar (configured via `pto_calendar_name` Config key, default `'Verizon Calendar'`). Classifies all-day events:

- Title contains `'Vacation'` → vacation day (pool: `pto_vacation_days`, default 20)
- Title contains `'PTO'` (all-day) → 8 personal hours (pool: `pto_personal_hours`, default 48)
- Title contains `'PTO'` (timed) → actual event duration in hours
- Title matches `pto_holiday_keywords` (default: `'Day,Holiday,Floating,Closure'`) → company holiday (not counted against PTO pools)
- Title matches `pto_ignore_keywords` (default: `'Pay Day'`) → skip entirely

Computes: days used, days remaining, ideal burn-rate pace (remaining days / remaining weeks in year), pace gap (actual vs ideal). Writes `[AUTO]` rows to the Summaries tab covering vacation balance, personal time balance, and pace status. Returns `ptoStats` object passed to Monthly Review.

Clear windows — `findClearWindows_()`

`findClearWindows_(gapCalendars, today, lookAheadDays, minWorkdays)` scans the next `lookAheadDays` days and finds contiguous stretches of `minWorkdays` or more clear workdays. "Clear" means no events on calendars defined by `pto_gap_calendars` Config key (default: `'Verizon Calendar,AE&VV - Our Joint Chaos'`).

Used by `getWeekendWindows()` in `WeekendPlanner.js` (filtered to windows touching a Friday or Monday) and by the Health Tracker to suggest appointment booking windows.

3-2-1 Framework

The PTO system surfaces a 3-2-1 planning framework in the dashboard PTO tab:

- **3 months out:** Plan big trips and time-off blocks
- **2 weeks out:** Confirm logistics, book details
- **1 day out:** Verify everything is ready

The `ptoSummaryForClaude()` function formats PTO data for injection into the Claude prompt, including 3-2-1 status indicators and a `paceStatus` field (`behind` , `on_track` , `ahead`).

12. Summaries Architecture

Two-tab architecture

Metrics tab (`TABS.METRICS`): VERA's self-monitoring. Written by `writeMetrics(ss)` . Contains task counts (open, overdue, neglected), calendar event counts (today, this week), and flag counts (total, by urgency). Fully automatic — no user configuration needed. Rows are `[AUTO]` -prefixed and rewritten every night.

Summaries tab (`TABS.SUMMARIES`): External life intelligence. Written by `writeSummaries(ss)` . Populated from external Google Sheets via `summary_sheet:SourceName` Config rows. Also receives `[AUTO]` rows from PTO snapshot, fitness data, and other module writes.

AUTO rows

Any row whose `Source` column starts with `[AUTO]` is owned by the script. The `clearAutoRows(sheet, tabName)` function:

1. Reads all rows in the sheet
2. Collects row indices where `Source` starts with `'[AUTO]'`
3. Deletes them from bottom to top (to preserve row indices during deletion)

After clearing, new `[AUTO]` rows are inserted before row 2 (just after the header) and styled with `background: '#eef2ff'` (subtle blue tint) and `fontColor: '#333333'` . Manual rows (without `[AUTO]` prefix) are never touched.

External sheet hooks

Config tab syntax:

Setting	Value
---------	-------

```
summary_sheet:SimpleAssTracker | SHEET_ID|TabName|CellRef|metric_name
```

For example:

```
summary_sheet:SimpleAssTracker | 1aBc...XyZ|Budget|B2|checking_balance
summary_sheet:SimpleAssTracker | 1aBc...XyZ|Budget|B5|groceries_actual_vs_budget
summary_sheet:Fitness          | 1xYz...AbC|Log|D4|gym_sessions_this_week
```

`writeSummaries()` reads all Config rows with keys starting with `summary_sheet:`. For each, it opens the external sheet via `SpreadsheetApp.openById(sheetId)`, reads the specified cell, and writes an `[AUTO]` row to the Summaries tab: `source = '[AUTO] SourceName'`, `metric = metric_name`, `value = cellValue`, `asOf = today`.

The external sheet must be shared (at minimum viewer access) with the Google account running the script (`aaeleraky@gmail.com`).

`getSummaries()`

Reads both the Metrics tab and the Summaries tab and returns a merged array of `{ source, metric, value, asOf }` objects. This merged array is what `generateFlags()` and chat context assembly receive — they do not distinguish between the two tabs.

13. Monthly Life Review

Trigger

`checkMonthlyReview_(ptoStats)` is called from `nightlyRun()` Step 0a. The function immediately returns if `today.getDate() !== 1` — it only runs on the 1st of each month. A second guard checks whether a review with the key `monthly_review_YYYYMM` (for the prior month) already exists anywhere in the Flags tab (any status) — if found, it returns without re-generating. This dedup ensures the review fires exactly once per month even if `nightlyRun()` somehow triggers multiple times on the 1st.

The Config key `monthly_review_enabled` (default `'true'`) gates the entire feature.

Assembly — `buildMonthlyReview_(label, priorMonth, ptoStats)`

Assembles the review by calling independent section builders, each wrapped in `try/catch`:

- **Goals:** Current-year goals grouped by status (Done, Doing, Stalled, Backlog)
- **Tasks:** Open task count, overdue count, most neglected tasks by age
- **Finance:** Reads spending data from the Summaries tab — `[AUTO]` rows with finance/SAT source
- **PTO:** Uses the `ptoStats` object passed from Step 0b (avoids re-reading the calendar)
- **Travel:** Reads the Itinerary tab for trips whose date range overlaps the prior month

- **Flags:** Counts flags generated in the prior month, resolved vs still-open
- **Carry Forward:** A single Claude call asking for one insight to carry forward from the month's data

Output

Three outputs are produced:

1. **Flag:** Written via `writeFlags()` with `source: 'Monthly Review'`, `urgency: 'Low'`, `key: 'monthly_review_YYYYMM'`. The full review text goes in the `reason` field.
2. **Archive:** Appended to the `Monthly Reviews` tab (headers: `Month Key | Month Label | Generated Date | Review`) as an append-only record. This tab is never modified by VERA — only appended.
3. **Email:** Sent via `sendMonthlyReviewEmail(label, reviewText)` as a styled HTML email.
4. **Slack log:** `sendSlackLog('📅 Monthly review written – March 2026')`.

14. Configuration Architecture

Config tab (runtime-tunable)

The `Config` tab in the Life OS sheet contains key-value pairs read by `getConfigValues()` on every execution:

```
function getConfigValues() {
  if (_configCache_) return _configCache_; // per-execution cache
  var ss = getSpreadsheet();
  var sheet = ss.getSheetByName(TABS.CONFIG);
  // ... reads all rows into an object ...
  _configCache_ = cfg;
  return cfg;
}
```

`_configCache_` is a module-level variable declared as `var _configCache_ = null` in `Code.js`. It resets automatically at the start of each Apps Script execution (there is no persistent memory between executions). Within a single execution (e.g. a nightly run), `getConfigValues()` is called 4–6 times — the cache prevents the Config tab from being re-read on every sub-function call.

Config keys that change behaviour have sensible defaults hard-coded as fallback strings (e.g. `cfg['calendar_days_ahead'] || '7'`). This means the system works out of the box with an empty Config tab; users only add rows to override defaults.

Config tab keys used across the system include:

Key	Default	Module	Purpose
-----	---------	--------	---------

<code>calendar_days_ahead</code>	<code>'7'</code>	Calendar.js	Look-ahead window for events
<code>task_age_threshold_days</code>	<code>'7'</code>	Tasks.js	Age before a task is "neglected"
<code>max_flags_per_night</code>	<code>'8'</code>	Claude.js (<code>CONFIG.MAX_FLAGS</code>)	Cap on Claude flag output
<code>skip_calendars</code>	<code>'Holidays in United States'</code>	Calendar.js	Calendars to exclude entirely
<code>calendar_label:CalName</code>	(none)	Calendar.js	Custom label for a calendar name
<code>pto_vacation_days</code>	<code>'20'</code>	PTO.js	Annual vacation allocation
<code>gym_sessions_per_week</code>	<code>'3'</code>	PatternRecognition.js, GymTracker.js	Weekly gym target
<code>reminders_enabled</code>	<code>'true'</code>	Reminders.js	Master switch for Anticipator
<code>explorer_enabled</code>	<code>'true'</code>	Reminders.js	Master switch for Explorer
<code>email_parser_enabled</code>	<code>'false'</code>	EmailParser.js	Master switch for inbox scanner
<code>pretrip_briefing_enabled</code>	<code>'true'</code>	PreTripBriefing.js	Enable/disable pre-trip flags
<code>monthly_review_enabled</code>	<code>'true'</code>	MonthlyReview.js	Enable/disable monthly review
<code>pattern_max_flags</code>	<code>'2'</code>	PatternRecognition.js	Max cross-domain flags per run
<code>pattern_dedup_days</code>	<code>'7'</code>	PatternRecognition.js	Dedup window for pattern flags
<code>pacing_enabled</code>	<code>'true'</code>	Pacing.js	Enable/disable pacing system
<code>pacing_flag_threshold</code>	<code>'3'</code>	Pacing.js	Unacked flags to trigger miss scoring
<code>summary_sheet:SourceName</code>	(none)	Summaries.js	External sheet hook definition

Script Properties (deployment secrets)

Read by `PropertiesService.getScriptProperties()` . These are set once in the Apps Script editor (Project Settings → Script Properties) and never committed to source control.

Property	Used by	Purpose
<code>VERA_SHEET_ID</code>	<code>Code.js</code> (<code>CONFIG.SHEET_ID</code>)	The Life OS Google Sheet ID
<code>MORNING_NUDGE_EMAIL</code>	<code>Code.js</code>	Email address for error notifications and email-mode nudges
<code>CLAUDE_API_KEY</code>	<code>Claude.js</code>	Anthropic API key
<code>SLACK_BOT_TOKEN</code>	<code>Slack.js</code>	Slack bot OAuth token
<code>SLACK_CHAT_CHANNEL_ID</code>	<code>Slack.js</code>	<code>#vera-chat</code> channel ID
<code>SLACK_NOTIFICATIONS_CHANNEL_ID</code>	<code>Slack.js</code>	<code>#vera-notifications</code> channel ID
<code>SLACK_LOGS_CHANNEL_ID</code>	<code>Slack.js</code>	<code>#vera-logs</code> channel ID
<code>SLACK_AHMED_USER_ID</code>	<code>Slack.js</code>	Ahmed's Slack user ID (for @mentions)
<code>SLACK_VICTORIA_USER_ID</code>	<code>Slack.js</code>	Victoria's Slack user ID
<code>SLACK_ALLOWED_USER_IDS</code>	<code>Slack.js</code>	Comma-separated allowlist of Slack user IDs
<code>AVIATIONSTACK_KEY</code>	<code>FlightStatus.js</code>	AviationStack API access key
<code>VERA_SEARCH_API_KEY</code>	<code>Chat.js</code>	Serper.dev or Tavily API key for web search
<code>VERA_SEARCH_ENGINE</code>	<code>Chat.js</code>	<code>'serper'</code> or <code>'tavily'</code>
<code>PACING_MODE_ACTIVE</code>	<code>Pacing.js</code>	Runtime state flag
<code>PACING_MODE_ENDS</code>	<code>Pacing.js</code>	Runtime state — expiry date
<code>VACATION_MODE_ACTIVE</code>	<code>Pacing.js</code>	Runtime state flag
<code>VACATION_MODE_ENDS</code>	<code>Pacing.js</code>	Runtime state — trip end date
<code>VACATION_TRIP_NAME</code>	<code>Pacing.js</code>	Runtime state — active trip key
<code>VERA_LOGO_FILE_ID</code>	<code>Code.js</code> (morningNudge)	Google Drive file ID for email logo

Decision rule for where a value lives

- **Secret or credential** (API key, channel ID, user ID, sensitive token) → Script Property. Never in the sheet, never in source control.
- **Value the user might want to tune** (thresholds, labels, calendar names, feature flags) → Config tab with a sensible default fallback in code.

- **Value that changes system behaviour** (feature on/off switches, interval days, target counts) → Config tab with a sensible default fallback. The fallback ensures the system works without any Config row being present for that key.
- **Value that is computed at runtime and needs to survive between executions** (pacing mode active/inactive, vacation mode, AviationStack backoff window) → Script Properties as ephemeral state. These are not secrets but they need to persist across the 6-minute execution boundary.